

Modelling How Savings Grow with Compound Interest

EconEdLink Compound Interest Calculator · EconEdLink

Years 7-10

~30 minutes

Mathematics · Number & Financial Maths

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://econedlink.org/resources/compound-interest-calculator/>

Curriculum links: AC9M7N06 · AC9M7N09 · AC9M10A04

Learning intentions

- I can use a compound interest calculator to model how savings grow over time.
- I can compare how changing the interest rate, deposit or time period affects total savings.
- I can use the results to justify a savings decision in a realistic financial context.

Getting started

Open the calculator and enter a starting deposit of your choice (e.g. \$500), an interest rate (e.g. 4%) and a monthly contribution (e.g. \$50). Run it for a starting time period of 5 years before changing anything else.

Tasks

1. Record your starting deposit, interest rate, monthly contribution and the final balance after 5 years.

2. Keeping everything else the same, change the time period to 10 years, then 20 years. What do you notice about how the balance grows — is it a steady, constant increase each year?

3. Keeping your deposit and time period (10 years) fixed, try three different interest rates and record the final balance for each.

Interest rate	Final balance after 10 years

4. Using your table, explain what happens to total savings when the interest rate increases, and why this effect gets bigger over a longer time period.

5. By hand (not the calculator), estimate the simple, non-compounding interest earned on the same deposit and rate over 10 years, and compare it to the calculator's compound result. Show your working.

Working:

6. Keep the interest rate and time period fixed, but double your monthly contribution. How much extra does this add to your final balance?

7. Two savers start with the same deposit and rate. One contributes for 5 years then stops, the other contributes for 15 years then stops — both leave their money invested after that. Use the calculator to compare their outcomes and explain why starting early matters, even with smaller deposits.

Extension (optional)

Use the calculator to work out roughly how long it would take to double an initial deposit at a fixed interest rate with no extra contributions, then research the 'Rule of 72' and compare it to what you found.
